

Ning Liang-bo

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Education

- Sep. 2023 – Present **Ph.D., Computing,** **GPA:3.49/4.3**
The Hong Kong Polytechnic University (PolyU), Hong Kong, China
Supervisor: Prof. Qing Li & Prof. Wenqi Fan
- Sep. 2020 – Jun. 2023 **M.E., Control Science and Engineering,** **GPA:88.00/100 or 3.75/4**
Northwestern Polytechnical University (NPU), Xi'an, China
Supervisor: Prof. Zhun-ga Liu
- Sep. 2016 – Jun. 2020 **B.E., Automation,** **GPA:83.47/100 or 3.34/4**
Northwestern Polytechnical University (NPU), Xi'an, China

Publications (First Author)

- 2026 **L. Ning**, W. Fan, Q. Li, "Retrieval-Augmented Purifier for Robust LLM-Empowered Recommendation." *Transactions on Information Systems (TOIS)*, 2025. (CCF-A, Online)
- 2025 **L. Ning**, Z. Liang, Z. Jiang, et al., "A Survey of WebAgents: Towards Next-Generation AI Agents for Web Automation with Large Foundation Models." *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '25)*, 2025. (CCF-A, Online)
- L. Ning**, W. Fan, Q. Li, "Exploring Backdoor Attack and Defense for LLM-empowered Recommendations." *arXiv*, 2025. (Preprint)
- L. Ning**, Z. Zhang, W. Ding, et al., "Multi-level Distribution Alignment for Multi-source Universal Domain Adaptation." *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2025. (JCR:Q1, IF:10.2, Online) [ESI Highly Cited Paper (2025 Nov.)]
- 2024 **L. Ning**, Z. Dai, W. Fan, et al., "Joint Universal Adversarial Perturbations with Interpretations," *arXiv*, 2024. (Preprint)
- L. Ning**, S. Wang, W. Fan, et al., "CheatAgent: Attacking LLM-Empowered Recommender Systems via LLM Agent," *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24)*, 2024. (CCF-A, Online)
- L. Ning**, Z. Dai, J. Su, et al., "Interpretation-Empowered Neural Cleanse for Backdoor Attacks," *The Web Conference (WWW '24)*, 2024. (CCF-A, Online)
- 2022 Z. Liu (**Master Supervisor**), **L. Ning**, and Z. Zhang, "A New Progressive Multi-Source Domain Adaptation Network with Weighted Decision Fusion," *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2022. (JCR:Q1, IF:10.2, Online)
- 2021 **L. Ning**, Z. Liu, and Z. Zhang, "Domain Adaptation Based Multi-Source Data Fusion for Pattern Classification," *CIE International Conference on Radar (Radar)*, 2021. (Online)

Publications (Collaboration)

- 2025 Y. Zhang, H. Qu, **L. Ning**, et al. "SSD4Rec: a structured state space duality model for efficient sequential recommendation." *Transactions on Information Systems (TOIS)*, 2024. (CCF-A, Online)
- Z. Zhang, Z. Liu, **L. Ning**, et al. "Belief-based fuzzy and imprecise clustering for arbitrary data distributions." *IEEE Transactions on Fuzzy Systems (TFS)*, 2025. (JCR:Q1, IF:10.8, Online)

Publications (Collaboration) (continued)

- 2024
- X. Yuan, **L. Ning**, W. Fan, and Q. Li. "mKG-RAG: Multimodal Knowledge Graph-Enhanced RAG for Visual Question Answering." *arXiv*, 2025. **(Preprint)**
 - J. Huang, S. Wang, **L. Ning**, et al., "Towards Next-Generation Recommender Systems: A Benchmark for Personalized Recommendation Assistant with LLMs." *ACM International Conference on Web Search and Data Mining (WSDM '25)*, 2025. **(CCF B, Online)**
- 2024
- H. Qu, **L. Ning**, R. An, W. Fan, et al., "A Survey of Mamba," *arXiv*, 2024. **(Preprint)**
 - Y. Ding, W. Fan, **L. Ning**, et al., "A survey on RAG meets LLMs: Towards retrieval-augmented large language models," *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24)*, 2024. **(CCF-A, Online)**
- 2023
- Z. Zhang, Z. Liu, **L. Ning**, et al., "Representation of Imprecision in Deep Neural Networks for Image Classification," *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2022. **(JCR:Q1, IF:10.2, Online)**
 - Z. Zhang, **L. Ning**, Z. Liu, et al., "Mining and Reasoning of Data Uncertainty-induced Imprecision in Deep Image Classification," *Information Fusion*, 2023. **(JCR:Q1, IF:10.2, Online)**
- 2022
- Z. Liu, Z. Zhang, Q. Pan, and **L. Ning**, "Unsupervised Change Detection From Heterogeneous Data Based on Image Translation," *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, vol. 60, pp. 1-13, 2022. **(JCR:Q1, IF:8.125, Online)**

Miscellaneous Experience

Awards and Achievements

- 2023
- **Excellent Master's Thesis**, NPU, Xi'an, China.
 - **Outstanding Graduates**, NPU, Xi'an, China.
- 2022
- **The National Scholarship**, Ministry of Education, China.
 - **The First Prize Scholarship**, NPU, Xi'an, China.
- 2019
- **The Second Prize Scholarship**, NPU, Xi'an, China.
- 2018
- **The First Prize Scholarship**, NPU, Xi'an, China.
 - **The Endress+Hauser Scholarship**, NPU, Xi'an, China.
- 2017
- **The Second Prize Scholarship**, NPU, Xi'an, China.

Certifications

- 2020
- **The First Prize of the 15-th National University Students Intelligent Car Race.**
- 2019
- **The First Prize of the 14-th National University Students Intelligent Car Race.**
- 2018
- **The Second Prize of the 13-th National University Students Intelligent Car Race.**
Awarded by Intelligent Car Race Organization Committee.

Skills and Research Interests

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|----------------|---|---|
| Coding | ■ | Pytorch, Matlab, C/C++, $\LaTeX$, Git. |
| Languages | ■ | Mandarin Chinese (Native), English (CET6, IELTS 6.5). |
| Res. Interests | ■ | Data Mining, WebAgents, Recommender Systems, Adversarial Attack, Domain Adaptation, Pattern Recognition, XAI. |